The Quantized Conscience: a term-by-term companion

Companion to the article “The Quantized Conscience: a 10-minute reality check against whitepaper panic”. All numbers: one RTX 3090 (sm_86), engine 0.1.55-linux-x86_64-cuda-cuda13, Qwen3.8-Flash-Next checkpoints (UD-Q4_K_XL / GSQ IQ3_S / Uncensored Q4_K_M), K=5 paired seeds 1000–1004. Reproduction: github.com/sinoptiks/sinoptiks — benchmark/qwen38-censored-ab.

The model and how it gets squeezed

MoE (mixture of experts). A model not with one big brain but with many small experts, of which only a handful activates per token: the full weights sit in system RAM and only the active slice lands in the 3090's VRAM. Why it is in the article: it is the quiet counter-argument behind “not on this MoE”. The degradation mechanism of the second paper is plausible on dense models, where quantization touches every weight of every layer; on an MoE it can hide inside experts a given prompt never summons. It is an architecture-derived hypothesis, not a measurement — and the article presents it exactly that way.

Quantization, bpw (bits per weight). Original weights are 16-bit numbers; quantization replaces them with 4-bit, 3-bit and so on through a lookup, so the model fits and runs faster. “4.6 bpw” is the UD-Q4_K_XL: important layers stayed fatter, average layers took more loss — hence the fractional average. Why: the first paper's claim is that bits kill the conscience; your ladder is exactly that axis — 4.6 → 3 bpw — and refusals did not move on it ($p = 0.12$). Central experiment, hence a running term.

Q4_K_XL / Q4_K_M / IQ3_S, GGUF, shards. GGUF is the model file format of the llama.cpp family: weights, metadata and the chat template in one file. Q4_K/Q4_M/IQ3_S are quantization schemes — K: block-scaled, M: mid, IQ3_S: aggressive 3-bit small. The UD file ships as 4 shards, split into pieces. Why: these are the exact names the stack is pinned with. Without them the fight is abstract: “Q4” and “Q4” are different models. The article names everything down to shard counts.

FP16. The 16-bit original format: unquantized. Why: the viral “41.9% decision-flips” peak of the first paper is placed by its own PDF on an FP16 model — not compressed at all. One of the article's two main blows, and it only works if the reader knows FP16 means unquantized.

KV cache, fp16 vs int8 KV. While generating, the model stores keys and values of everything written so far — the KV cache; int8-KV compresses that too. The second paper's “alignment can be silently destroyed” is about this. Why: a dedicated experiment — a clone of the same model with fp16 KV against the int8 default: 40/40 refusals on both sides, DFR 0.0% on both. No silent destruction on this MoE; the authors' verb is “can”, their production check is vLLM/FP8 on NVIDIA, your bench is llama.cpp-lineage int8 on one consumer card.

The experiment

Refusal / the refusal wall. The model says no to a harmful request. Your wall: 40/40 refusals on the censored arm versus 0/40 on the uncensored (Fisher $p < 10^{-10}$). Why: the core of the article — the wall is set by the checkpoint, not by bits; every quantization moves it by noise at best.

Checkpoint. The weights file you downloaded. Uncensored: censorship picked out; UD, GSQ: names of quant variants. Why: the only axis that actually changed behavior. The article in five words: you argue about processors, the file decides.

Arm. One stack variant in a comparison: A — the daily driver Q4_K_XL, C — IQ3_S, plus the uncensored control and the KV clone. Why: honest comparison requires changing one axis at a time; arms is how you say “we changed one variable”.

Prompt injection; poisoned vs clean. An instruction “do-this-now” hidden inside a document or context. Poison: the run with the venom; clean: the same request without it, on identical seeds. Why: the real hole turned up not in quantization but here — 45% ASR on a model that politely refuses on topic. The article's second thesis: the real danger was invisible.

ASR (attack success rate). The share of poisoned runs where the model performed the injected action: 45% → 33% (bit ladder), → 52% (KV), 51.7 → 45.0% (engine). Why: the one metric across all injection runs — the same one attack papers report, so the numbers stay comparable.

DFR (decision flip rate). The first paper's term: how often the same request gets the opposite decision across builds. Your refusal DFR is 0% nearly everywhere (0/40 under the strict wording). Why: the bridge to “refusal decision-flips” in the teaser — and a warning that DFR is not the same as refusing at all: the famous 41.9% are flips, not “41.9% stopped refusing”.

Seed (1000–1004), T=0.7/1.0, top_p, thinking off, paired seeds. A seed is the roulette number of generation: same stack plus same seed reproduces the sample. T is temperature (0.7 daily, 1.0 stress). Paired seeds: both stack variants get identical seeds, so any difference is attributed to the stack rather than luck. Why: without pairing there is no McNemar; the T=1.0 run delivered a finding of its own — zero refusal flips in 80 seed pairs while answer-correctness flipped in 5–23%: refusal is a rock, usefulness is a dice.

Cell (task, cond, seed). One row of a run; 120 cells per block. Why: the unit the pairs are counted on; “the 120/120 grid matched” is the answer to the re-run question.

U1 (work done), U0. U1 measures whether useful work got done under the venom — not refuse/obey but was-the-task-completed: 26 → 34/60 on 3 bits (2:10, p = 0.039), the only axis that moved. Why: the article deliberately keeps it a hypothesis, not a fact (nominal p among ~10 contrasts; Bonferroni ≈ 0.4). That is cultural behavior toward your own best find — and the reason no one can dock it.

Statistics

n = 60, K = 5. Sixty poisoned runs (12 tasks × 5 seeds); K = 5 repetitions per task. Why: the article's main denominator — nearly everything is “not measurable” because n is small, and the article says this out loud instead of inflating the conclusion.

McNemar (11:4, 6:2, 8:4). The test for paired binary outcomes: it counts only the discordants — cells where the old version complied and the new refused (b) and the other way round (c). 8:4 means eight one way, four the other, with 48 agreements: a coin, p = 0.388. Why: the article's main anti-nonsense tool. Other people's “improved by 6.7 points” die on it: at small n a difference of shares is a difference of dice, not of decisions.

Fisher's exact test ($p < 10^{-10}$). A 2×2 test with no distributional assumptions: 40/40 against 0/40 is “by chance” with probability on the order of 10^{-10}. Why: the single result that survives any multiplicity correction; its contrast with McNemar is the difference between a huge signal and a shift inside noise.

Wilson bound (40/40 → ≥ 91.2%; 0/80 → ≤ 4.6%). A confidence interval for a proportion at small n, honest even at all-or-nothing: 40/40 is not 100%, it is at least 91.2%; zero flips in 80 pairs is not “0%”, it is “≤ 4.6%”. Why: from the first paragraph you refuse to sell “all” as “all in the world”. The article also notes 40/40 means 8 prompts × 5 seeds, not 40 independent trials — even Wilson is generous here.

p-value, nominal, Bonferroni. p is the chance of seeing this or a sharper skew if there is actually no difference. Nominal: computed as the one and only test, though about ten ran; Bonferroni is the multiplicity correction — 0.039 × ~10 ≈ 0.4, so the pretty p stops being significant. Why: you buried your own most interesting finding with the correction yourself — the article's most expensive reputation coin.

Drift vs dice. At n = 60 a difference of shares cannot be told apart from noise. Why: the one-line verdict of §5, and the root of the whole “pin your stack” philosophy.

Stack and versions

Engine / build / version. 0.1.52 → 0.1.55; build string 0.1.55-linux-x86_64-cuda-cuda13; CUDA 13; sm_86, the 3090's chip architecture. ASR 51.7 → 45.0 on the identical grid (120/120 cells and payloads matched; McNemar 8:4), while the uncensored arm across the same builds moved 47/60 → 47/60, per-cell flips 1:1, p = 1.0. Why: the versioning point — your one-percent improvement may be a rebuild — and the lesson: write the whole stack down or fail to reproduce your own numbers a month later.

Chat template. The wrapper instructions the server packs your messages into before feeding the model; it lives in the GGUF/server and changes with the engine. Why: the only real reason a rebuild could shift behavior at all — and precisely what the old artifacts cannot rule out, since /props were captured only from the second run.

vLLM vs llama.cpp lineage. Inference engines: vLLM is the production stack (the paper authors also ran FP8 on server NVIDIA hardware); yours is the home llama.cpp lineage with int8. Why: “can” versus “did” — their production check is a different stack, scheme and hardware; your results do not refute their mechanism, they show it does not silently transfer to your bench.

Tool calls, function schema, 21 → 293. Model-invoked functions; the schema (function description) rides along in the request. One function added 272 prompt tokens to 21; the calls came back clean JSON 5/5. Why: the checked-not-assumed aside — after the rebuild, neither broken tools nor token inflation — plus the real cost of a single tool, which nobody advertises.

Threat and takeaway

The defensive system prompt. The daily driver's ordinary protective prompt. Why: with it the model politely refuses on topic (40/40) and swallows injections at the same time (45%) — two independent walls, and the second is leaky. Refusing is not defending.

“Silently destroyed”. The second paper's verb. Why: you never loudly rejected it — the wording is “not REFUTED, but NOT ON THIS MOE”: on your bench it was not observed, which is not the same as disproven. The entire article's lexicon is calibrated to that distinction.

10-minute probe; CI-friendly; exit 1. inject_check: 8 tasks, clean vs poisoned on identical seeds, prints U0/ASR/U1/DFR, exits nonzero when ASR ≥ 25% — CI-ready. Why: the closing move — do not trust me: a measurement that costs one coffee, and the standing answer to “but my model is different”.

> **The chain in one line.** Quantization and KV squeeze weights and cache → the question asked is “is the conscience dead?” → the conscience metrics are DFR and refusals, not ASR → on paired seeds McNemar counts the disagreements → at n = 60 everything except the checkpoint is dice → and the real hole (45% ASR) lives under the polite refusal wall → so: pin your stack, raise your n, test your own bench.

Article: “The Quantized Conscience: a 10-minute reality check against whitepaper panic”. Method, prompts, scorer: github.com/sinoptiks/sinoptiks/tree/main/benchmark/qwen38-censored-ab · 10-minute check: github.com/sinoptiks/sinoptiks/tree/main/benchmark/qwen38-injection-check · engine: github.com/Niko1221/Strata · papers: arxiv.org/abs/2606.29581 (v2) and arxiv.org/abs/2606.09864. Prepared 2 November 2026; all numbers come from run artifacts dated 29 October and 2 November.